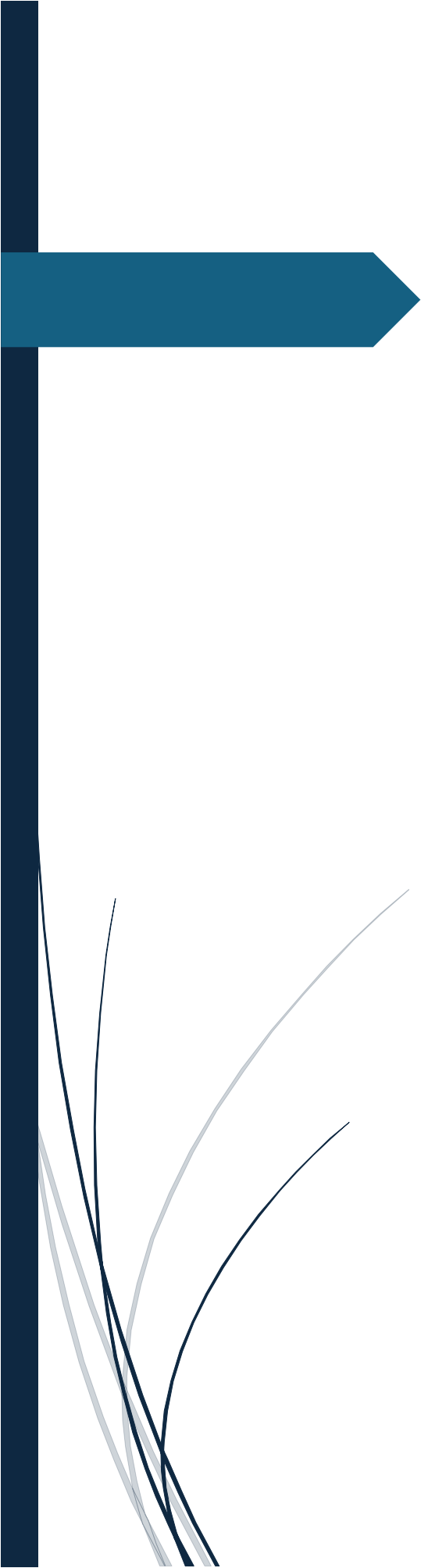
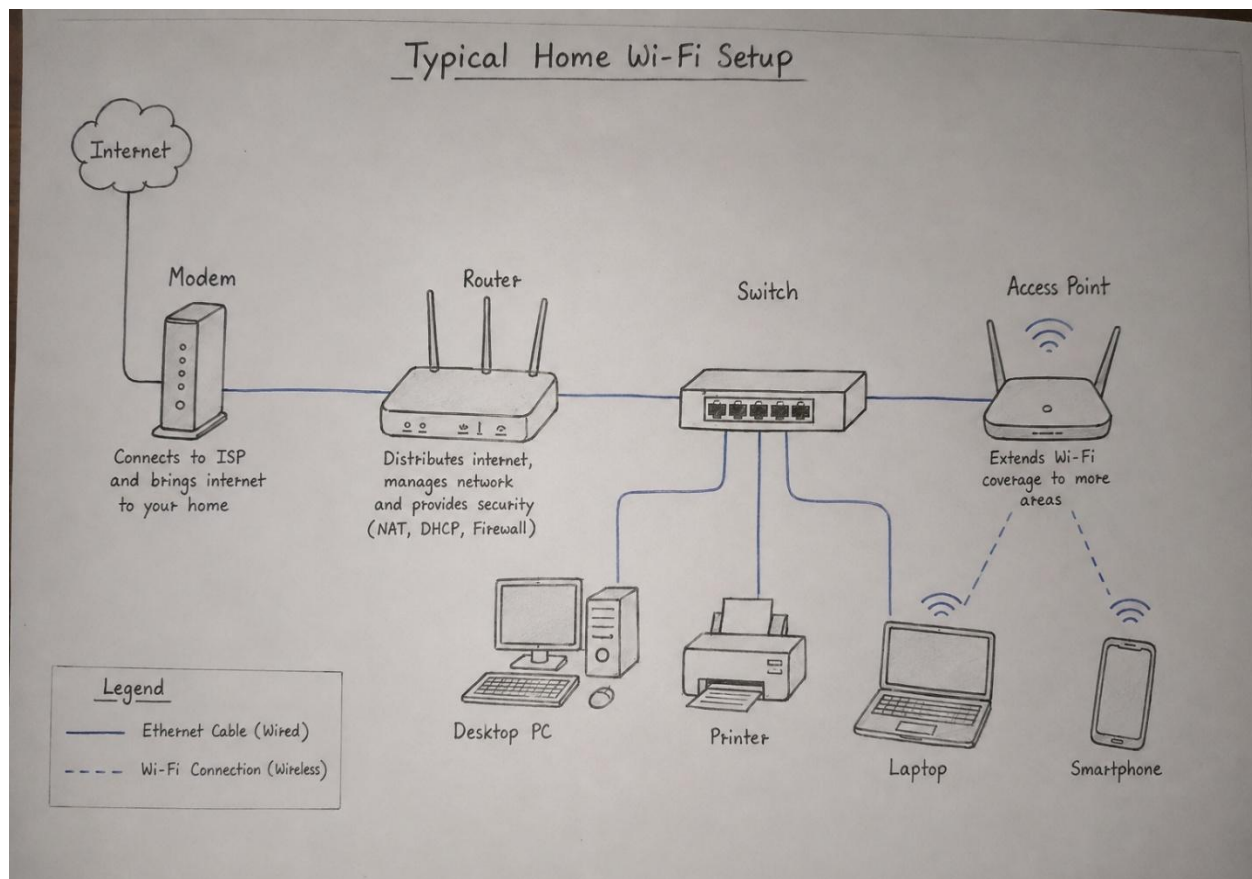




Assignment 1

Introduction of Networking

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Device	Main Function	Real-Life Example
Modem	Connects your home to the Internet.	Used at home to receive Internet from the ISP.
Router	Shares the Internet with many devices and manages the network.	Used at home to connect phones, laptops, and TVs.
Switch	Connects multiple devices using network cables.	Used in an office to connect computers and printers.
Access Point	Provides Wi-Fi to devices.	Used in a school to provide Wi-Fi in classrooms.
Network Cable	Carries data between network devices.	Ethernet cable used to connect a PC to a router.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

➤ **Ethernet cable**



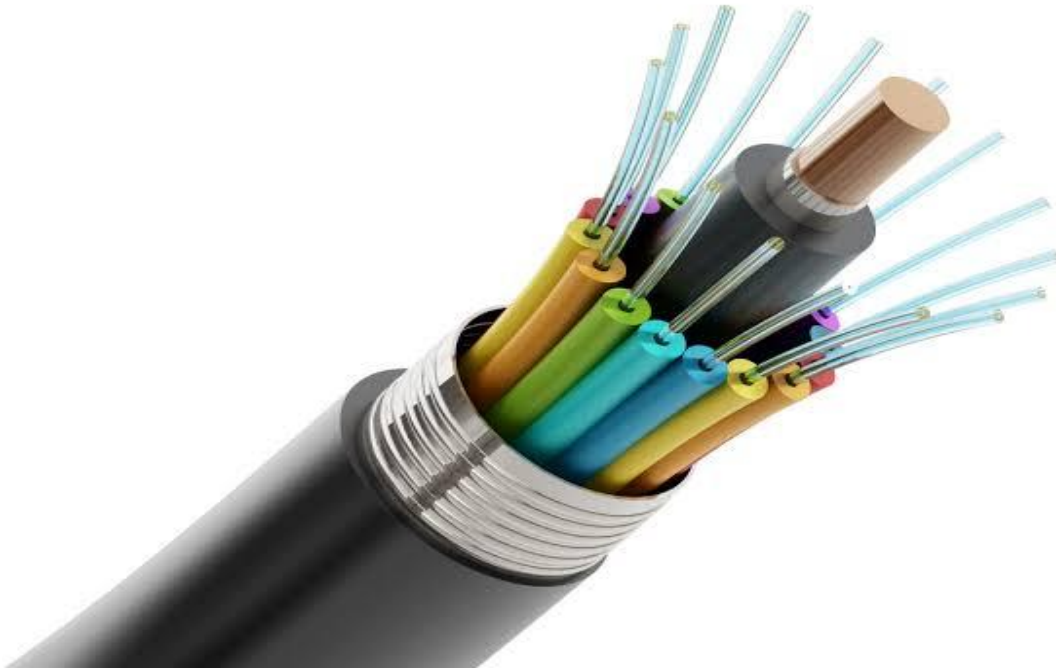
Ethernet Cable: Commonly used to connect computers, printers, and routers inside a local area network (LAN).

➤ **Coaxial cable**



Coaxial Cable: Commonly used to deliver cable television signals and broadband internet to homes.

➤ Fiber optic cable



Fiber Optic Cable: Commonly used for high-speed, long-distance data and internet communication.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main use	Connects your home/office network to the internet	Connects devices (like computers, printers) to each other in the same room or building
Where it's used	At the "edge" — between your network and the outside world	Inside your network, connecting many devices together
How it sends data	Uses IP addresses (like a home address) to send data to the right network	Uses MAC addresses (like a device's ID number) to send data to the right device
Number of ports	Few ports — usually one for internet, a few for devices	Many ports — can connect lots of devices at once
Simple way to think about it	Like a mailman who sends letters between different cities	Like an office receptionist who passes messages between people in the same building